Tonmya_Prescriptions

2026-03-29

Some simple models based on Tonmya sales

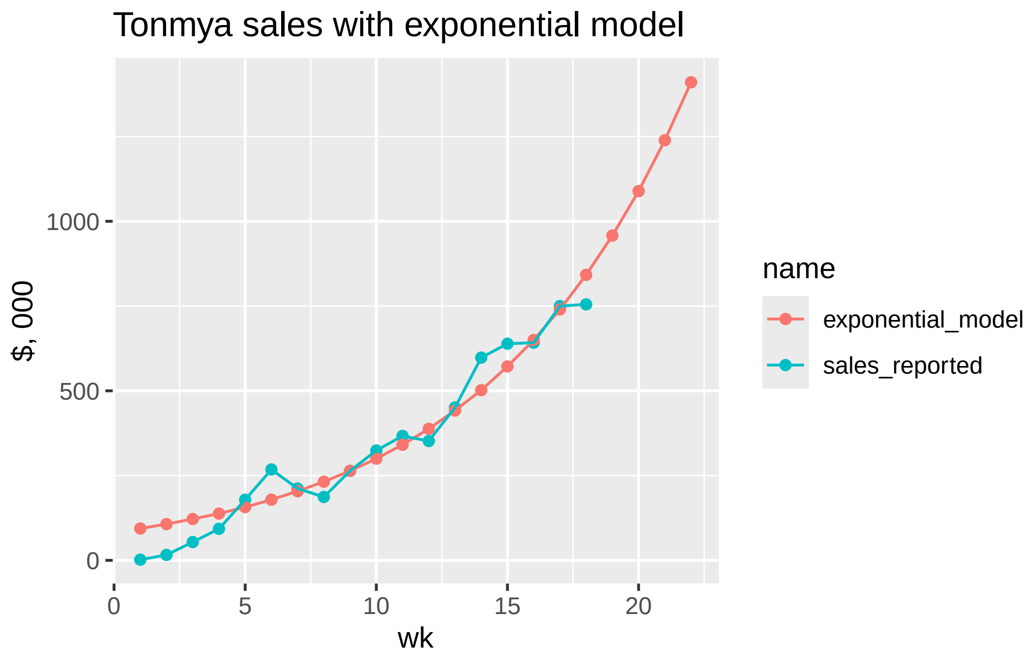
What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

Sales Projection - at weekly growth rate - *exponential model*

```
Formula: sales ~ a * (1 + r)^wk

Parameters:
  Estimate Std. Error t value Pr(>|t|)
a 82.53419   12.49188   6.607 6.03e-06 ***
r  0.13770    0.01135  12.131 1.76e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.1 on 16 degrees of freedom

Number of iterations to convergence: 13
Achieved convergence tolerance: 1.543e-06
```

Exponential Model

wk	sales, 000	exp, 000	annual, M
1	$2	$94	$4.9
2	$16	$107	$5.6
3	$54	$122	$6.3
4	$93	$138	$7.2
5	$179	$157	$8.2
6	$268	$179	$9.3
7	$212	$204	$10.6
8	$187	$232	$12.0
9	$264	$264	$13.7
10	$324	$300	$15.6
11	$367	$341	$17.7
12	$352	$388	$20.2
13	$451	$442	$23.0
14	$598	$502	$26.1
15	$639	$572	$29.7
16	$642	$650	$33.8
17	$750	$740	$38.5
18	$755	$842	$43.8
19	NA	$958	$49.8
20	NA	$1,089	$56.7
21	NA	$1,239	$64.5
22	NA	$1,410	$73.3

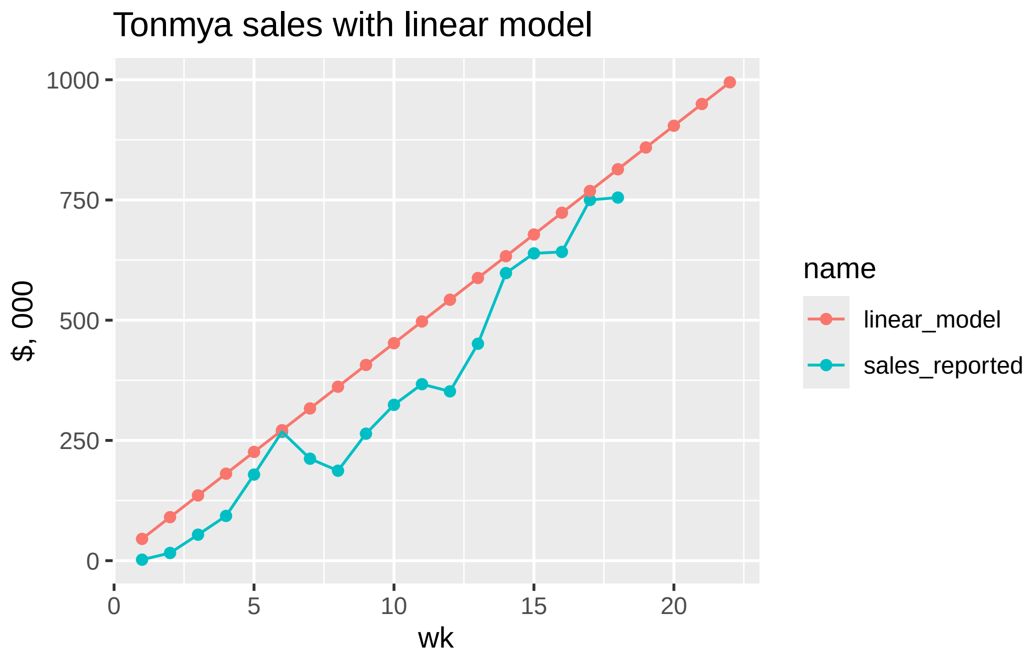


Sales Projection - at weekly rate - *linear model*

```
Call:
lm(formula = sales ~ wk, data = df_sales)

Residuals:
     Min       1Q   Median       3Q      Max
-102.870  -42.102    6.167   43.527   84.418

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -87.706     27.238   -3.22  0.00535 **
wk            45.215      2.516   17.97 4.96e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55.39 on 16 degrees of freedom
Multiple R-squared:  0.9528,    Adjusted R-squared:  0.9498
F-statistic: 322.8 on 1 and 16 DF,  p-value: 4.958e-12
```

Linear Model

wk	sales, 000	exp, 000	annual, M
1	$2	$45	$2.4
2	$16	$90	$4.7
3	$54	$136	$7.1
4	$93	$181	$9.4
5	$179	$226	$11.8
6	$268	$271	$14.1
7	$212	$317	$16.5
8	$187	$362	$18.8
9	$264	$407	$21.2
10	$324	$452	$23.5
11	$367	$497	$25.9
12	$352	$543	$28.2
13	$451	$588	$30.6
14	$598	$633	$32.9
15	$639	$678	$35.3
16	$642	$723	$37.6
17	$750	$769	$40.0
18	$755	$814	$42.3
19	NA	$859	$44.7
20	NA	$904	$47.0
21	NA	$950	$49.4
22	NA	$995	$51.7

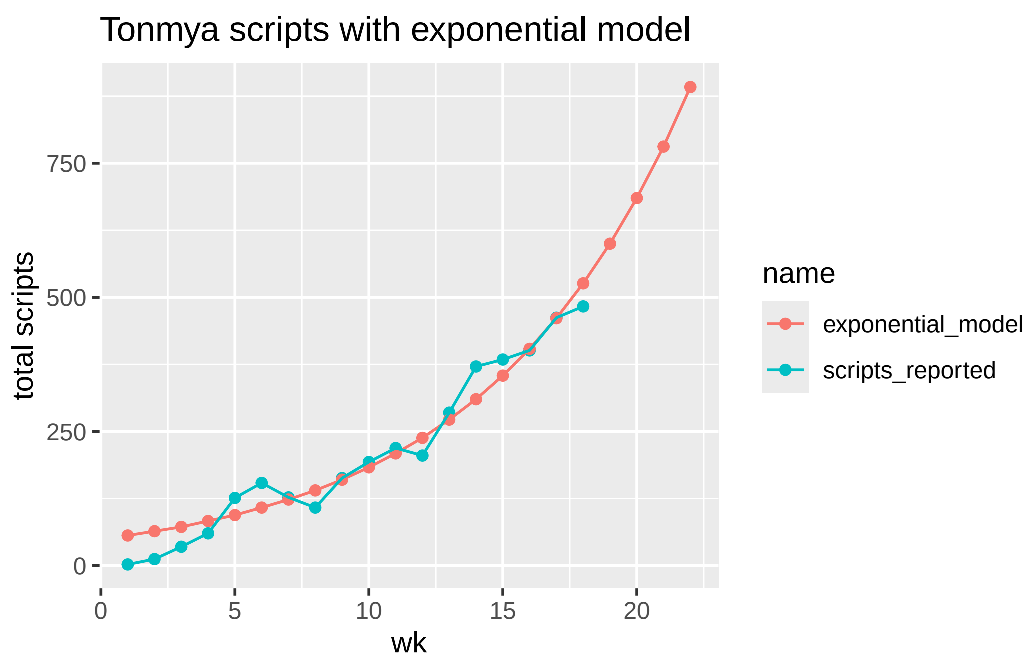


Scripts Projection - at weekly growth rate - *exponential model*

```
Formula: scripts ~ a * (1 + r)^wk

Parameters:
  Estimate Std. Error t value Pr(>|t|)
a 48.77611    7.08852   6.881 3.69e-06 ***
r  0.14121    0.01089  12.966 6.65e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 35.02 on 16 degrees of freedom

Number of iterations to convergence: 8
Achieved convergence tolerance: 6.409e-06
```

Scripts, Exponential Model

wk	total reported, 000	exp predicted, 000	annual, M
1	2	56	2.9
2	12	64	3.3
3	35	72	3.8
4	60	83	4.3
5	126	94	4.9
6	154	108	5.6
7	127	123	6.4
8	108	140	7.3
9	163	160	8.3
10	193	183	9.5
11	219	209	10.8
12	205	238	12.4
13	285	272	14.1
14	371	310	16.1
15	384	354	18.4
16	401	404	21.0
17	462	461	24.0
18	483	526	27.3
19	NA	600	31.2
20	NA	685	35.6
21	NA	781	40.6
22	NA	892	46.4

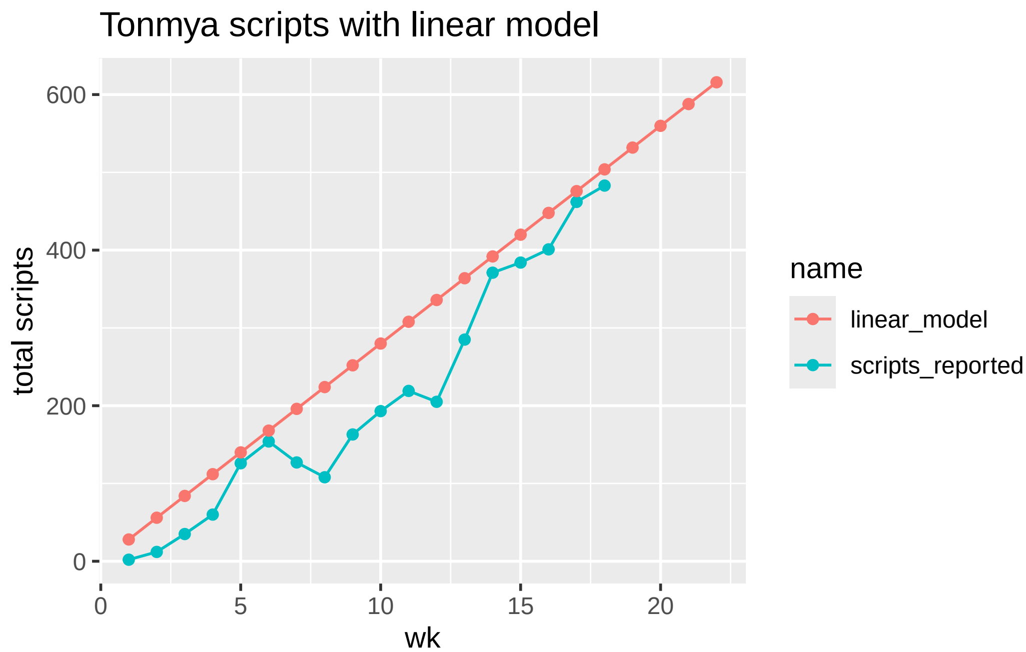


scripts Projection - at weekly rate - *linear model*

```
Call:
lm(formula = scripts ~ wk, data = df_scripts)

Residuals:
    Min      1Q  Median      3Q     Max
-75.535 -29.545   7.444  33.205  41.506

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -55.366     18.395   -3.01  0.00831 **
wk            27.992      1.699   16.47 1.86e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 37.41 on 16 degrees of freedom
Multiple R-squared:  0.9443,    Adjusted R-squared:  0.9408
F-statistic: 271.3 on 1 and 16 DF,  p-value: 1.864e-11
```

Scripts, Linear Model

wk	total reported, 000	linear predicted, 000	annual, M
1	2	28.0	1.5
2	12	56.0	2.9
3	35	84.0	4.4
4	60	112.0	5.8
5	126	140.0	7.3
6	154	168.0	8.7
7	127	195.9	10.2
8	108	223.9	11.6
9	163	251.9	13.1
10	193	279.9	14.6
11	219	307.9	16.0
12	205	335.9	17.5
13	285	363.9	18.9
14	371	391.9	20.4
15	384	419.9	21.8
16	401	447.9	23.3
17	462	475.9	24.7
18	483	503.9	26.2
19	NA	531.8	27.7
20	NA	559.8	29.1
21	NA	587.8	30.6
22	NA	615.8	32.0



Tonmya scripts with linear model
total scripts
600
400
200
0
0
5
10
15
20
wk
name
linear_model
scripts_reported